

## Coding & Solution

|                      |                 |
|----------------------|-----------------|
| <b>Date</b>          | 1 July 2026     |
| <b>Team ID</b>       | SWTID-2026-8096 |
| <b>Project Name</b>  | Rising Waters   |
| <b>Maximum Marks</b> | 5 Marks         |

## Solution Summary

| Field                         | Details                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | <a href="https://github.com/Jagadeesh-krishna/Flood-Prediction-/tree/main">https://github.com/Jagadeesh-krishna/Flood-Prediction-/tree/main</a>                                                                                                                                                                                                                                                 |
| Programming Language(s)       | Python, HTML, CSS                                                                                                                                                                                                                                                                                                                                                                               |
| Framework(s) Used             | Flask, Scikit-learn, Pandas, NumPy                                                                                                                                                                                                                                                                                                                                                              |
| Key Features Implemented      | <ul style="list-style-type: none"> <li>• Flood prediction using Machine Learning</li> <li>• User-friendly web interface</li> <li>• Rainfall data input form</li> <li>• Flood/No Flood prediction result page</li> <li>• Responsive design with animated rainfall background</li> <li>• Trained Random Forest model integration</li> </ul>                                                       |
| Pending / Incomplete Features | <ul style="list-style-type: none"> <li>• User authentication</li> <li>• Historical prediction records</li> <li>• Live rainfall data integration through APIs</li> <li>• Interactive flood visualization maps</li> </ul>                                                                                                                                                                         |
| Setup / Run Instructions      | <ol style="list-style-type: none"> <li>1. Clone the GitHub repository.</li> <li>2. Install dependencies using pip install -r requirements.txt.</li> <li>3. Run the application using python app.py.</li> <li>4. Open the local server URL (or deployed Render URL) in a web browser.</li> <li>5. Enter rainfall values and click <b>Predict</b> to view the flood prediction result.</li> </ol> |

## Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

## Additional Notes / Comments

- The project follows a modular Flask application structure for better readability and maintainability.
- A trained Random Forest Machine Learning model is used to predict flood occurrence based on rainfall inputs.
- The application has been successfully deployed and tested.
- Git is used for version control, enabling efficient code management and collaboration.
- The user interface includes a responsive layout with a rainfall animation to enhance user experience